

The empirical recurrence for OEIS A237237: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A237237 records “Empirical recurrence of order 67”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 914$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 67, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 67 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A237237 is “Number of $(n+1)X(3+1)$ 0..3 arrays with the maximum plus the lower median minus the upper median minus the minimum of every $2X2$ subblock equal.”. It has offset 1 and begins

6976, 91744, 1301304, 19274172, 289676104, 4449131420, 68828263944, 1081186864940, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:00:32, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 914$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 914$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1828$ exact terms of the model returns order 67 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	51	c_{18}	−60605990815134138779	c_{35}	−86420276294025151012385131
c_2	137	c_{19}	3567387139845518241561	c_{36}	−654554840789484150778688320
c_3	−48611	c_{20}	−3152590398318447324918	c_{37}	9873694247676362506996075720
c_4	549558	c_{21}	−124004894784300769968238	c_{38}	3510054024419418377941153224
c_5	17200606	c_{22}	268378911175025176041477	c_{39}	−748705964058970480168252374
c_6	−349905123	c_{23}	3239401069909686635360001	c_{40}	−1392718820385814965410811127
c_7	−2434100311	c_{24}	−10740286976525527010130487	c_{41}	4141285724357429287247717888
c_8	97715133711	c_{25}	−63483707926055181525231475	c_{42}	3776845456450326512560937894
c_9	−57773857461	c_{26}	290412431364133550654823406	c_{43}	−1721276330908046314051079344
c_{10}	−15545770508508	c_{27}	912550808064578520446816934	c_{44}	−5079501579174143549829479388
c_{11}	71815078507556	c_{28}	−5781025047388370272165411368	c_{45}	54135829042083571248469676391
c_{12}	1551785956098856	c_{29}	−8934942668431318598393829608	c_{46}	−8230379391879349954595295343
c_{13}	−12140659623890632	c_{30}	87605663910790291443373577360	c_{47}	−12802520846440694349995488231
c_{14}	−100114158769184690	c_{31}	42590239431122456140446871056	c_{48}	65541631575301672144652617979
c_{15}	1169987685071013190	c_{32}	−1025537962434857438004297080704	c_{49}	223306704476426939201109594492
c_{16}	3954623619345977257	c_{33}	301054815676928700772190355840	c_{50}	−18844811309621309104292735355
c_{17}	−76146246758240216715	c_{34}	9316440685657744607670775102848	c_{51}	−27610370151866962520952941144

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 67, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $67 + 4$ terms removed returns order 67 as well, so $\deg q_{\min} = 67 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $67 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 67 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A237237.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.